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PROFESSIONAL SUMMARY

High-achieving Computer Engineering undergraduate at the GUC with international research experience at **Universität Augsburg**. Specializes in **Applied Machine Learning**, **Algorithmic Optimization**, and **Full Stack Development**. Proven track record of academic excellence (Category “A” Scholar) with peer-reviewed research in **wearable-based affective computing** and **multi-agent systems optimization**.

EDUCATION

- **B.Sc. in Computer Engineering** (Media Engineering & Technology) | **German University in Cairo (GUC)** | 2021 – 2026
 - **Bachelor Thesis:** "Integration of Wearable Sensors into Breathing-based Emotion Regulation Application Interface" – Completed during a semester exchange at **Universität Augsburg, Germany (Grade: A+)**.
 - **Cumulative GPA:** 2.15 (German Scale).
 - **Awards:** Classified as Category "A" Student (Awarded 25% Scholarship).
- **British High School Diploma (IGCSE)** | **Al-Bashaer International School** | **Grade: 121.25%** | *Graduated 2021*

RESEARCH & WORK EXPERIENCE

- **Wearable-Based Real-Time Stress Detection and Breathing Intervention** | *EAI PervasiveHealth 2025 (in press)*
 - **Lead author.** Built an end-to-end ECG-based stress detection system using a **Random Forest** classifier (81% LOSO accuracy on WESAD) integrated with a non-intrusive guided breathing intervention via **React Native / Flask**.
- **Optimization of Multi-AGV Scheduling for Airport Baggage Handling** | *ICEENG-IPCS 2026 (in press)*
 - **Co-author.** Benchmarked SA, GA, ACO, Q-Learning, and a proposed **Hybrid ABC** algorithm on a **VRPTW** formulation; ABC achieved **61%** efficiency gain over FCFS baseline with **40%** fleet reduction.
- **IT Networks and Applications Intern** | National Automotive Company (NATCO) | *June 2024 – Aug 2024*
 - Developed an Employee Records Management System using **C# .NET** and **EF Core**, implementing full CRUD operations.
 - Designed normalized SQL schemas and utilized **LINQ** queries to optimize data retrieval and reporting performance.

PROJECTS

- **Intelligent Airline Assistant & Knowledge Graph** | *Python, Neo4j, Scikit-Learn*
 - **Predictive Modeling:** Developed a pipeline to predict passenger satisfaction using Random Forest/FFNN with SHAP/LIME explainability.
 - **Knowledge Graph RAG:** Architected a Neo4j RAG system to answer complex travel queries using vector embeddings and graph traversal.
- **Multi-Agent Cooperative Systems Optimization** | *Python, Matplotlib*
 - Solved the Vehicle Routing Problem (VRPTW) for AGVs using meta-heuristics (GA, SA, ACO, ABC) to minimize travel costs under capacity constraints.
- **AI Game Agents & Logic Reasoning** | *Java, Prolog*
 - Engineered an autonomous Minimax agent with Alpha-Beta Pruning for battle strategy and developed a Prolog planning agent using Situation Calculus to solve complex logic problems.
- **Custom Database Management System** | *Java*
 - Engineered a high-performance database engine from scratch, implementing B+ Tree indexing, page serialization, and binary search algorithms to optimize CRUD operations.
- **Smart Mirror System** | *Raspberry Pi Pico, React, C*
 - Designed an embedded system featuring dual TFT screens and ultrasonic sensors, developing a React-based interface for real-time news display, LED control, and buzzer alerts.
- **Virtual Trip Planner** | *MERN Stack (MongoDB, Express, React, Node.js)*
 - Architected a full-stack web application for travel budgeting and itinerary planning, integrating third-party APIs for real-time booking data and secure user authentication.

VOLUNTARY WORK & ACTIVITIES

- **Competitive Swimmer:** Dedicated **15+ years** to competitive swimming, achieving multiple medals in national and regional competitions while maintaining a rigorous training schedule alongside full-time engineering studies.

SKILLS

- **Programming:** Java, Python, C, SQL, Haskell, Prolog, TypeScript.
- **Web & Frameworks:** React Native, Node.js, Express.js, Flask, JavaFX, .NET, Entity Framework, REST APIs.
- **Tools & Frameworks:** Scikit-learn, Arduino, Raspberry Pi, VHDL, Git/GitHub, JUnit, LaTeX.
- **Languages:** Arabic (Native), English (Fluent), German (B1/Intermediate)